

Problem Statement 02: Oil Spill Detection

Background

Major maritime oil tanker casualties release large volumes of heavy crude, creating environmental disasters that threaten vulnerable shorelines, marine sanctuaries, port infrastructure, and coastal desalination intakes.

Problem

Oil slick transport is governed by complex physical dynamics (Fay spreading, surface current advection, wind drift, and evaporation weathering). Spill response commanders have limited physical containment resources (skimmers, booms, dispersant aircraft) and must deploy them under strict environmental exclusion zones prohibiting chemical dispersants near coastal fringes.

Aim

Develop a model that detects oil spills from GNSS-R Delay-Doppler Map (DDM) images and estimates the oil film thickness.

Output

1. Complete, reproducible training and inference source code (train.py, predict.py)
2. Exported model weights/checkpoints (.pt or .onnx)
3. Predictions CSV mapping sample_id to oil presence probability and thickness

Data

File Structure:

PS2_public/

├── train/images/

| ├── img_001.png

| ├── img_002.png

| └── ...

└── test/images/

```
| |— img_501.png
| |— img_502.png
| |— ...
|— train_labels.csv
|— sample_submission.csv
```

Data Description:

Images:

1. **Format:** PNG (grayscale)
2. **Size:** 400 × 400 pixels
3. **Type:** GNSS-R Delay-Doppler Maps (DDM)
4. **Physical meaning:** Oil spills dampen sea surface roughness, altering GPS signal reflection patterns

train_labels.csv:

Column	Description	Type
sample_id	Unique image identifier	int
image_path	Relative path to image	string
oil_present	Is there oil? (0/1)	int
oil_thickness_mm	Oil thickness (mm)	float

Note: Labels are only in train_labels.csv

Requirements

1. The submission file must contain exactly 3 columns: `sample_id`, `oil_present`, and `oil_thickness_mm`.
2. `oil_present` must be a float between 0 and 1, and `oil_thickness_mm` must be a float greater than or equal to 0.0. The file must contain no missing values (NaN).
3. The submission must contain exactly 250 rows, with `sample_id` matching the test image filenames, and must be saved as `teamname_PS2.csv`.

Example Submission

sample_id	oil_present	oil_thickness_mm
image_001	0	0
image_002	1	0.20
image_003	1	4.2

Evaluation Metrics

Your submission will be scored on:

1. LogLoss (Classification): Lower is better (70% weight).
2. RMSE (Regression): Lower is better (30% weight).

Final Score = $(0.7 \times \text{LogLoss}) + (0.3 \times \text{RMSE})$

sample_id,oil_present,oil_thickness_mm

image_001,0,0
image_002,1,0.20
image_003,1,4.2